

Name: Pa Pichsakanya

Class: Afternoon

Assignment: Answer 4 Questions

1. Why Consensus Exists (Centralized vs Decentralized)

In traditional centralized systems, trust is simple. One organization owns the system, controls the database, and has the authority to decide what information is correct. For example, when a bank updates your account balance, everyone accepts it because the bank is trusted to maintain accurate records. Even if mistakes happen, there is a clear entity responsible for fixing them. Because of this structure, centralized systems do not need consensus among many parties only the central authority decides. However, decentralized systems work very differently. There is no single owner or controller. Instead, many independent participants (nodes) maintain their own copy of the data. These participants may be in different countries, may not trust each other, and may even act maliciously. Without a central authority, there must be a mechanism to ensure that everyone agrees on the same version of history. This is why consensus exists. Consensus allows a decentralized network to function reliably by ensuring that honest participants reach agreement even when some participants fail, act slowly, or try to cheat. In short, consensus replaces trust in an institution with trust in rules and mathematics.

2. Safety, Liveness, and Finality

These three properties describe whether a consensus protocol works correctly and usefully over time.

2.1. Safety

Safety means the system never reaches an incorrect state. Once a transaction is accepted as valid, it should never be contradicted later by another conflicting transaction. For example, if Alice sends money to Bob, safety ensures that the same money cannot also be sent to someone else. Safety is critical because without it, users cannot trust the system's results.

2.2. Liveness

Liveness means the system continues to make progress. Transactions submitted to the network will eventually be processed and confirmed. Even if some nodes crash, disconnect, or behave maliciously, the system should not stop working. A system that is safe but does not confirm transactions is technically correct but practically useless.

2.3. Finality

Finality refers to the point at which a decision becomes irreversible. Once a transaction reaches finality, it is considered permanent and cannot be undone. Some systems provide strong finality, where a transaction is final immediately after confirmation. Others provide probabilistic finality, where confidence increases over time as more blocks are added.

3. Consensus in P2P Networks

In a peer-to-peer (P2P) network, every node plays an equal role. There is no central server that distributes or verifies information. Instead, nodes communicate directly with one another, forming a network graph where information spreads from node to node. When a transaction or block is created, it is broadcast to neighboring nodes. Each node independently verifies the data according to the protocol rules. Through repeated communication and validation, the network gradually converges on a shared state. Even though nodes may receive messages at different times or in different orders, consensus ensures they eventually agree on the same history. This decentralized topology improves resilience because the system does not depend on a single point of control or failure.

4. Common Attack Models

Consensus protocols are designed under the assumption that attackers exist and will attempt to exploit the system. One common attack is double spending, where an attacker tries to spend the same digital asset more than once. Consensus prevents this by enforcing a single agreed-upon transaction history. Another attack is the 51% attack, where an attacker gains control of the majority of network resources and attempts to manipulate the consensus outcome. While possible in theory, it becomes increasingly difficult as the network grows. Sybil attacks involve creating many fake identities to influence the network. Consensus mechanisms defend against this by tying influence to scarce resources such as computing power or economic stake.